UNIVERSITY | BUSINESS × COMPUTING

From workflows to autonomous agents

A practical map of the AI landscape — before building with n8n and Hermes Agent

30 MIN
Landscape

150 MIN
n8n workflow lab

150 MIN
Hermes agent lab

LANDSCAPE

Four layers. Four different engineering jobs.

“AI agent” is not one product category.

01 Research / search

Retrieve and synthesize evidence; agent-like multi-step search may be available.

ChatGPT Deep Research, Grok search

02 Workflow automation

Predefined triggers, rules, approvals and API actions.

n8n, Dify, Flowise

03 Agent SDK / orchestration

Develop loops, tools, state, handoffs and traceable behaviors.

OpenAI Agents SDK, Google ADK, LangGraph

04 Autonomous agent product

Delegate a goal; product manages multi-step action over time.

Hermes, Manus, Grok Bot, Meta Muse

Sources: provided agentic-platform report + official product documentation

ARCHITECTURE

What makes an agent an agent?

An LLM gets controlled authority to choose and call tools in a loop.

GOAL / USER

Risk-review an invoice

→

MODEL / PLANNER

Chooses next action

→

TOOLS / MCP

Read file · calculate · search

STATE / MEMORY

What happened before

GOVERNANCE

Permissions · approval

TRACE / EVAL

Audit every action

CRITICAL BOUNDARY An agent can call a tool ≠ an agent is authorized to cause a real-world effect

Sources: provided agentic-platform report + official product documentation

CONCEPTUAL TEST

Search is evidence. An agent is execution.

The same user goal can be approached with very different control structures.

Deep research / web search

Collect sources → compare claims → synthesize findings.

Useful for: competitor research, due diligence, market signals.

Risk: citation errors, freshness and source ambiguity.

Tool-using / autonomous agent

Plans work → selects tools → observes results → revises → stops.

Useful for: multi-step investigations, structured analysis.

Risk: unauthorized side effects and unreproducible decisions.

Teacher question: Which steps should never be left to the model?

Sources: provided agentic-platform report + official product documentation

LOW-CODE / VISUAL

Workflow builders: make business rules visible

The strongest default for predictable, auditable process paths.

FORM → Data capture

CODE → Validate rules

TABLE → Audit evidence

TELEGRAM → Notify humans

n8n — our Lab 1

Branching, Data Tables, notifications and Gemini explanation.

Other visual options

Dify, Flowise, enterprise Copilot Studio.

Sources: provided agentic-platform report + official product documentation

SDKs & frameworks: build the agent runtime

Different primitives for the same fundamental engineering problems.

OpenAI Agents SDK

Agent loop · tools · handoffs · tracing

Google ADK

Tool use · evaluation · multi-agent flows

Claude Agent SDK

Coding and task-execution agent patterns

Microsoft Agent Framework

Enterprise workflows and middleware

LangGraph

Explicit state, persistence and branching

CrewAI / Letta

Agent roles / long-term memory

Sources: provided agentic-platform report + official product documentation

Autonomous products: delegate a result

Products differ greatly in how much internal control students can inspect.

HERMES AGENT

Open, inspectable runtime

Tools, memory, skills, subagents, code execution.

GROK BOT

Persistent AI teammates

Own cloud computer; ongoing tasks and app access.

MANUS

Hosted task-oriented agent

Research, web tasks and end-to-end deliverables.

META MUSE

Managed personal agent

Product/VM-based personal action; access varies.

Sources: provided report

NEW PRODUCT · 2026

Grok Bot is not the Grok Agent Tools API

One is a persistent agent product; the other is a developer tool surface.

Grok Bot | launched Aug 11, 2026

AI teammates with their own computers, persistent conversations and delegated work.

Subsequently expanded to additional plans; X and enterprise integrations announced.

Product-level autonomy; desktop/iOS availability depends on plan.

Grok + Agent Tools API

Developer API for model/tool workflows, including search and code tooling.

You build orchestration, validation, storage and application controls.

It is not interchangeable with the Grok Bot application.

Primary sources: x.ai/news/grok-bot-launch · x.ai/news/designing-grok-bot

Same invoices. Two architectures.

What is gained when we move from a fixed workflow to a goal-driven agent?

Lab 1 · n8n (150 min)

TRIGGER → Normalize → Code Rules → Data Table → Telegram → optional Gemini.

Strength: predictable paths and deterministic controls.

Deliverable: classified invoices and execution evidence.

Lab 2 · Hermes (150 min)

GOAL → Explore files → Invoke verified finance tool → Memo → reusable skill → evaluate.

Strength: flexible investigation and procedural reuse.

Deliverable: CFO memo, actual trace and evaluation log.

Boundaries first: what students MAY do

Safe experiments are more valuable than dazzling but unauditable demos.

ALLOWED

- ✓ Read synthetic company CSVs
- ✓ Compute ratios with a deterministic tool
- ✓ Write a draft CFO memo
- ✓ Create reusable SKILL.md instructions
- ✓ Propose a human approval request

DISALLOWED

- × Move funds or approve a payment
- × Access personal/home folders or secrets
- × Connect production ERP/bank systems
- × Treat documents as system instructions
- × Assume model prose is a verified calculation

EVALUATION

Score the system, not the eloquence

Use the same criteria across n8n and Hermes.

Accuracy

Ratios and invoice classifications

Provenance

Which CSV, date, units and tool?

Tool behavior

Actual calls, not claimed calls

Controls

Never pay; human approval preserved

Efficiency

Elapsed time, tokens, intervention

Repeatability

Same result across repeated runs

TAKEAWAY

Your challenge: become an AI systems designer

The best workflow is not always the most autonomous agent.

Where should the rule be deterministic, and where does agent discretion create measurable value?

NEXT · n8n

Build reliable data and exception handling.

THEN · Hermes

Build a tool-verified, repeatable analysis skill.

REFERENCE LINKS

Verified sources and further reading

Primary documentation + supplied 2026 platform comparison.

Grok Bot introduction (Aug 2026)	https://x.ai/news/introducing-grok-bot
Grok Bot persistent design	https://x.ai/news/designing-grok-bot
Hermes Agent quickstart	https://hermes-agent.nousresearch.com/docs/getting-started/quickstart/
Hermes security model	https://hermes-agent.nousresearch.com/docs/user-guide/security
n8n documentation	https://docs.n8n.io/
OpenAI Agents SDK	https://openai.github.io/openai-agents-python/
Google ADK	https://google.github.io/adk-docs/